

Module 5 overview: recording script

INFO 521 · Machine Learning Foundations · Week 5 · instructor script

Target: 3 minutes. **Draft:** 321 words, about 2.5 minutes at 130 words per minute.

Slide headings below match the deck exactly. Advance on the heading.

Title slide

Module 5: Approximate Inference and Bayesian Classification

Module 5. This is the busiest week of the term, and it is also where the course stops having exact answers.

Slide: The question

(49 words, about 23 seconds)

The question is what you do when the posterior has no closed form. Binarise the outcome, put a logistic link on the linear model, and the conjugacy from last week is gone. There is no formula left to write down. So we need honest ways to proceed without one.

Slide: What the lectures do

(94 words, about 43 seconds)

Lecture A builds the logistic model and fits it with Newton-Raphson, which uses the curvature of the log-likelihood to get there in a handful of steps rather than crawling downhill. Lecture B gives you two approximations. Laplace fits a Gaussian at the mode, which is fast and often good enough. Metropolis draws samples from the posterior without ever computing the normalising constant, because in a ratio the constant cancels. That cancellation is the whole trick, and it is worth sitting with, because it is why sampling works at all on posteriors nobody can normalise.

Slide: Where this sits

(31 words, about 14 seconds)

This is the week the material starts looking like practice rather than algebra. It is also where the second half of the project opens, on the binarised outcome this module introduces.

Slide: Week 5 at a glance

(70 words, about 32 seconds)

Look at the week honestly before it starts. Homework Unit 5, checkpoint quiz 2.1 on the Newton update, Milestones 1.3 and 1.4 due, and the project midpoint: Part 1 due. Part 2 begins immediately. There is no discussion this week because of the midpoint. Starting this week on the Wednesday will not work. Look at what Part 1 asks for now, while you still have room to move things around.

Slide: Watch out for

(56 words, about 26 seconds)

Laplace only sees the peak. It is a Gaussian fitted at the mode, so skew and multimodality are invisible to it. That is fine right up until it is not, and Lecture B shows you a case where it is not. Knowing which of your approximations is lying to you is most of the skill here.